

# Lecture 11 Team Projects

## Interest-Rate Options

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### 1 Project 1: Cap, Floor, or Swap for a Borrower?

**Scenario.** Protect a floating-rate borrower against rising rates using a cap, swap, or no hedge.

**Objectives.** Compare payoff asymmetry, premium, opportunity cost, accounting, and liquidity; evaluate four rate scenarios; recommend within a risk budget.

**Sources.** CME SOFR information; instructor cap/swap quotes; New York Fed or FRED SOFR; public floating-rate debt disclosure.

**Deliverables.** Exposure profile, payoff table, scenario costs, volatility discussion, recommendation and trigger, and a 10–12 minute presentation.

**Rubric (100 points).** Setup 20; payoffs 30; option-versus-swap reasoning 20; recommendation 20; presentation 10.

## 2 Project 2: Option on Treasury Futures Event Hedge

**Scenario.** Limit downside around a policy or inflation release without surrendering all upside.

**Objectives.** Choose a futures option or spread; map payoff to duration exposure; compare with a linear futures hedge.

**Sources.** CME Treasury futures/options data; Federal Reserve, BLS, or Bank of Canada calendars; instructor option chain.

**Deliverables.** Event definition, contract and strike, payoff diagram, premium-inclusive scenarios, hedge comparison, and a 10–12 minute presentation.

**Rubric (100 points).** Framing 15; selection 25; payoff calculations 25; hedge judgment 25; presentation 10.

## 3 Project 3: Collar a Floating-Rate Liability

**Scenario.** Design a cap-and-floor collar for a borrower that wants a bounded interest rate at limited net premium.

**Objectives.** Choose cap and floor strikes; calculate all-in borrowing costs across scenarios; explain premium, basis, counterparty, and opportunity-cost tradeoffs.

**Sources.** CME SOFR futures/options; New York Fed SOFR; instructor cap/floor premiums and loan terms.

**Deliverables.** Instrument terms, payoff chart, six-scenario cost table, comparison with cap-only and swap alternatives, and a 10–12 minute presentation.

**Rubric (100 points).** Structure design 25; payoff accuracy 25; comparison 20; risk and recommendation 20; presentation and documentation 10.

## 4 Project 4: Volatility and Option Value Briefing

**Scenario.** Explain why two otherwise similar rate options have different premiums and how a volatility change affects a hedge budget.

**Objectives.** Compare intrinsic and time value; explain expiry, moneyness, volatility, and rates; test low/base/high volatility quotes; recommend a structure.

**Sources.** CME contract specifications and delayed quotes; instructor volatility and premium snapshots; FRED or U.S. Treasury rates.

**Deliverables.** Option comparison, premium decomposition, three-volatility sensitivity, budget tradeoff, recommendation, and a 10–12 minute presentation.

**Rubric (100 points).** Instrument comparison 20; value-driver analysis 25; sensitivity 25; recommendation 20; presentation and sources 10.